MIL-HDBK-738

29 August 1967

MILITARY STANDARDIZATION HANDBOOK

CLEANING OF OFFICE MACHINES

FSC 7930

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DEPARTMENT OF DEFENSE
WASHINTON, D. C. 20301

MIL-HDBK-738
Cleaning of Office Machines

1. This standardization handbook was developed by the Department of Defense.

2. This publication was approved 29 August 1967 for printing and inclusion in the Military Standardization Handbook series.

3. This handbook provides information and guidance to personnel preparing technical manuals, technical bulletins and other documents for field use. The handbook is not intended to be referenced in purchase specifications, nor shall it supersede any specification requirements.

4. Every effort has been made to reference the latest information on the cleaning of office machines. It is the intent to revise this handbook periodically to insure its completeness and currency. Users of this document are encouraged to report any errors discovered and any recommendations for changes or inclusions to the U. S. Army Natick Laboratories, Natick, Massachusetts 01760.

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FOREWORD

The intent of this handbook is to provide fundamental guidelines for the cleaning of office machines. This handbook concerns both first echelon cleaning and major tank cleaning performed in Military maintenance shops. These guidelines are intended to aid in the preparation of technical manuals, technical bulletins and other documents for field use.

Readers are cautioned that this handbook is not intended for procurement purposes and that information on commercial items is neither complete nor an indorsement of these products.

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1. INTRODUCTION

Office machines considered in this handbook include typewriters, teletypewriters, desk calculators, and adding machines. This handbook was not limited to cleaning supply items which would normally be in FSC Class 7930-Cleaning and Polishing Compounds and Preparations, as many DOD Activities are using such items in FSC Class 6810-Chemicals and FSC Class 7510-Office Supplies. The lubrication of office machines and the selection of shop equipment are also covered in this handbook because they are an integral part of overall cleaning operations.

The Department of Defense spends large sums of money for cleaning office machines. This equipment represents a sizeable investment and failure to give it proper care could result in excessive repair and replacement costs. By the same token, excessive cleaning or the misuse of cleaning products could also shorten the useful life of machines.

Numerous Military and commercial sources of information were selected to determine commonly used and recommended procedures and supplies for cleaning machines. The information which follows represents the best recommendations that could be developed from a large cross section of experience and opinions.

2. SELECTION OF ITEMS FOR CLEANING AND LUBRICATING OFFICE MACHINES

2.1 Selection of Cleaning Material Items

Table I gives a selection of cleaning material items reported to perform satisfactorily for cleaning office machines and a list of items to be avoided. All of the Federal Stock Numbers specified for these cleaning items are listed in the Federal Supply Catalogs applicable to the Defense General Supply Center, Richmond, Va.

Discretion should be used in reviewing the items listed for the entire tank cleaning of office machines. These are not necessarily the only ones that give satisfactory results. This list was obtained from replies to questionnaires sent to a representative cross section of Military installations and therefore is not conclusive of all the items that are available. Commercial facilities also verified the satisfactory performance of some of these items. However, it should be understood that individual users may have preferences based on availability from local suppliers, product form, odor, flammability due to solvents and previous experience with product effectiveness.

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2.2 Selection of Lubricating Material Items

Table II gives a selection of items recommended for lubricating office machines. These data were developed in the same manner as for Table I. The dry cleaning solvents listed in this table are intended to thin the oil, thereby allowing for better penetration into the minuscular spaces of the office machines. They are never to be used alone.

2.3 Selection of Equipment Items

In order to permit effective cleaning and lubrication, repair shops should be properly equipped with the items listed in Table III.

3. CLEANING AND LUBRICATING PROCEDURES

First echelon maintenance is essential to reduce frequent major tank cleaning and to increase the serviceable life of office machines. Personnel should make sure all office machines are covered when not in use to prevent dust and dirt from getting into the internal mechanical parts. In the case of typewriters, the carriage should be moved to one side prior to making an erasure to prevent bits of paper and eraser from dropping into the machine.

Tables IV, V, VI and VII give the proper procedures and frequencies for cleaning manual typewriters, electric typewriters, teletypewriters, and desk calculators, respectively.

There are 5 primary cleaning systems (cleaner/rinse) available which are pertinent to the cleaning material items in Table I. These are:

detergent-water cleaner/water rinse
detergent-solvent-water cleaner/water rinse
detergent-solvent cleaner/water rinse
detergent-solvent cleaner/solvent rinse
detergent-solvent cleaner/solvent-oil rinse

Further variations to these 5 cleaning systems are encountered when the different types of cleaning equipment are considered. These are:

pump or propellor agitated tanks
tanks in which the machine is agitated in the cleaner
tanks in which the machine is showered with the cleaner
air agitated tanks
soaking tanks
cleaning with a pressure spray hose
ultrasonic cleaning

All repair shops that do a moderate to large amount of cleaning should have some type of tank cleaning machine. Those with a very large amount of machine cleaning should investigate ultrasonic cleaning processes.

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TABLE I

SELECTION OF CLEANING MATERIAL ITEMS FOR OFFICE MACHINES

Cleaning Task	Cleaning Items Reported to Perform Satisfactorily	Cleaning Items to be Avoided
Type face of typewriters	FSN 7510-286-2220, Brush, Typewriter; 5 in. long FSN 7510-550-8446, Brush, Typewriter; 6 in. long FSN 7510-550-8448, Brush, Typewriter; 11 3/4 in. long FSN 7510-527-1458, Type Cleaner; liquid form* FSN 6810-292-9625, 1, 1, 1-Trichloroethane, technical*	FSN 7510-753-4768, Type Cleaner; sheet form FSN 7510-779-5720, Type Cleaner; sheet form FSN 7510-285-1745, Type Cleaner; plastic form FSN 7510-616-9588, Type Cleaner; liquid form; 6 oz. pressurized can Alcohol, Denatured
Rubber platens of typewriters	FSN 6810-201-0904, Alcohol, Denatured; 55 gal. drum* FSN 6810-201-0906, Alcohol, Denatured; 1 pt. bottle* FSN 6810-543-7415, Alcohol, Denatured; 1 gal. can* FSN 6810-543-7694, Alcohol, Denatured; 1 qt. bottle* FSN 6810-853-3034, Alcohol, Denatured; 4 oz. bottle*	1,1,1 - Trichloroethane, Technical
Entire typewriters, teletypewriters and desk calculators	Products For Use With Manufacturer Recommended Organic Solvent: 1/ +Ke Sonic C +Lix Mulsirex Actusol +Super Agitene +Magnusol Tally-Ho +Oakite No. 9 +Mulsine	Carbon tetrachloride (Note: Dangerously toxic) Gasoline (Note: Dangerously flammable and toxic)

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TABLE I (continued)

SELECTION OF CLEANING MATERIAL ITEMS FOR OFFICE MACHINES

Cleaning Task	Cleaning Items Reported to Perform Satisfactorily	Cleaning Items to be Avoided
Entire typewriters, teletypewriters and desk calculators (continued)	Products For Use With Water and Manufacturer Recommended Organic Solvent: 1/ Magnusol+ Actusol+ Mulsine+ Products For Use With Water: Chem-Mist DJ Cleaner Speed Soap Crystal Radient Economist Protexol Steamite Globrite Gunk Navee 42+	

* These items should be authorized for use by typewriter repairmen only.
\+ Agitation is not necessary.
1/ This is not endorsement of any of these commercial products, as this list is certainly not complete and some of the products may be more effective than others.

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TABLE II

SELECTION OF LUBRICATING MATERIAL ITEMS FOR OFFICE MACHINES

Lubricating Task	Recommended Lubricating Oils and Solvents	Lubricating Items to be Avoided
Typewriters, teletype-writers and desk calculators	FSN 9150-231-6689, Lubricating Oil, General Purpose; 1 qt. can FSN 9150-252-6173, Lubricating Oil, General Purpose; 4 oz. can FSN 9150-252-6174, Lubricating Oil, General Purpose; 1 qt. can FSN 9150-231-9045, Lubricating Oil, General Purpose; 1 gal. can FSN 6850-264-9037, Dry Cleaning Solvent; 55 gal. drum* FSN 6850-264-9038, Dry Cleaning Solvent; 5 gal. pail* FSN 6850-274-5421, Dry Cleaning Solvent; 5 gal. pail* FSN 6850-285-8011, Dry Cleaning Solvent; 55 gal. drum*	Heavy weight oils

* These items are used to thin oil and are never used alone.

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TABLE III

RECOMMENDED EQUIPMENT ITEMS FOR CLEANING AND LUBRICATING OFFICE MACHINES

Two tanks (automatic or manual), one for washing and one for rinsing.

Compressed air with a pressure of approximately 70 p.s.i.

A spray gun.

Hot water with a minimum temperature of 165°F.

Safety containers for storing solvents.

A drying oven with a temperature of approximately 212°F. (Note: The drying oven should be ventilated in accordance with NBFU Pamphlet No. 86, Standards of the National Board of Fire Underwriters for Class A Ovens and Furnaces as recommended by the National Fire Protection Association, November 1950, National Board of Fire Underwriters, New York, New York). Drying ovens are unfeasible for shiptender use because repair shops on board are quite small.

A spray hood with an exhaust fan for use in repair shops where machines are lubricated by spraying with a solvent and oil mixture. (Note: The spray hood should be enclosed on all sides but one and exhaust ventilated outdoors. In addition, the rate of exhaust should be sufficient to provide an indraft of 100 f. p. m. or higher in accordance with Table I, Page 10, "American Standard Safety Code for the Design, Construction, and Ventilation of Spray Finishing Operations", Z9.3-1964, American Standard Association, Inc., 10 East 40th Street, New York, New York 10016).

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TABLE IV

PROPER USE OF CLEANING AND LUBRICATING ITEMS FOR MANUAL TYPEWRITERS

Cleaning Task	Procedure	Frequency
Type face	A sheet of paper should be placed under the typewriter keys to catch pieces of dirt, ink or other matter while the type face is vigorously brushed until clean.	Close of each working day.
Carriage	The carriage should be brushed to remove bits of erasers and paper that accumulate throughout the day.	Daily or more frequently, if needed.
Rubber platens*	The rubber platens should be cleaned by wiping with a rag dampened with denatured ethyl alcohol.	At least once per year. Otherwise, after each service call.
Entire typewriter*	Preparation for Cleaning: Several parts on the typewriter must be removed prior to immersion in the solution tank. This may vary with the make and model, but generally these include all rubber parts, i.e., platens, bail rolls, feed rolls, and feet, and all cover plates that may shake loose during agitation. Dust Removal: After removal of all the necessary parts, the typewriter should be carefully blown out with air. Too much air pressure may loosen or damage the internal springs. Cleaning Solution: The typewriter should then be placed on a rack and lowered into the solution tank where either the typewriter or the solution is agitated for 10 to 30 minutes, depending on the type of cleaning solution and dirt accumulation.	Once each year.

* To be cleaned by typewriter repairmen only.

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TABLE IV (continued)

PROPER USE OF CLEANING AND LUBRICATING ITEMS FOR MANUAL TYPEWRITERS

Cleaning Task	Procedure	Frequency
Entire typewriters (continued)	Agitating the typewriter in the solution is more effective than agitating the solution around the typewriter. A pressure spray hose coming directly from the solution tank is very effective for penetrating impervious spaces. Draining and Drying: The typewriter should then be removed from the solution, allowed to drain for about 15 minutes, and rinsed out thoroughly with a stream of hot water. Immediately following this rinse, the excess water should then be blown off with air and the typewriter should then be oven dried. Extra care should be taken to make certain that the typewriter is blown completely dry and immediately lubricated when a drying oven is not available. Lubricating: Upon removal from the drying oven, the typewriter should be lubricated with a spray gun containing approximately 1 part light oil and 6 parts dry cleaning solvent or soaked in a solution consisting of approximately 1 part light oil and 5 parts dry cleaning solvent. Then the typewriter should be blown with air to remove the solvent and excess oil.	

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TABLE V

PROPER USE OF CLEANING AND LUBRICATING ITEMS FOR ELECTRIC TYPEWRITERS

Cleaning Task	Procedure	Frequency
Type face	Same as for manual typewriters.	Close of each working day.
Carriage	Same as for manual typewriters.	Daily or more frequently, if needed.
Rubber platens*	Same as for manual typewriters.	At least once each year. Otherwise, after each service call.
Entire typewriters*	Same as for manual typewriters except that the motor, the line cord and other electrical parts should be removed prior to immersion in the solution.	At least once every 1 1/2 years.

TABLE VI

PROPER USE OF CLEANING AND LUBRICATING ITEMS FOR TELETYPEWRITERS

Cleaning Task	Procedure	Frequency
Entire teletypewriters*	Same as for electric typewriters, except that these machines should not be rinsed with water because of their more complex internal structure. Instead, these machines should be placed on a drain board and blown completely dry with air.	At least once every 1 1/2 years.

TABLE VII

PROPER USE OF CLEANING AND LUBRICATING ITEMS FOR DESK CALCULATORS

Cleaning Task	Procedure	Frequency
Entire desk calculators*	Same as for electric typewriters except lubrication should be done by soaking the machine in the oil and solvent bath rather than spraying with the oil and solvent solution.	At least once every 1 1/2 years.

* To be cleaned by typewriter repairmen only.

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Project No. 7930-0226

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